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Quiz 3

The most basic and important way to judge: Target policy = Behavior policy or not .

1 SARSA :

On-policy Learning

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)]$$

- The update uses the actual next action A' , which is selected by the agent's current behavior policy (e.g., ϵ -greedy) in state s' , not the optimal action.
- The policy being learned is the same policy that generates data.

2 Q-learning :

Off-policy Learning

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

- The update uses the maximum Q-value of the next state s' (i.e., $\max_{a'} Q(s', a')$), which corresponds to the value of the optimal action for s' .
- It does not use the actual next action A' taken by the agent.
- The target policy (greedy optimal policy) is different from the behavior policy (ϵ -greedy, used for exploration).